NVIDIA Corporation

Launches Next Gen Data Center Platform; Strong Software/Ecosystem Development To Maintain High Barriers To Entry; Reit. OW

Overweight

NVDA, NVDA US
Price (18 Mar 24):$884.55

Price Target (Dec-24):$850.00

Semiconductors & Semiconductor Capital Equipment / IT Hardware

Harlan Sur AC
(1-415) 315-6700
harlan.sur@jpmorgan.com

Peter K Peng
(1-415) 315-8327
peter.k.peng@jpmchase.com

Pritesh Patel
(1-415) 315-3933
pritesh.n.patel@jpmorgan.com
J.P. Morgan Securities LLC

Quarterly Forecasts (FYE Jan)

Adj. EPS ($)

	2024A	2025E	2026E
Q1	1.09	5.32	
Q2	2.70	5.49	
Q3	4.02	5.80	
Q4	5.16	6.09	
FY	12.96	22.70	

Style Exposure

Quant Factors	Current %Rank	Hist %Rank (1=Top) 6M	1Y	3Y	5Y
Value	81	88	84	77	79
Growth	25	80	71	74	33
Momentum	5	1	46	55	93
Quality	2	7	6	7	2
Low Vol	58	63	63	45	72
ESGQ	21	9	11	2	1

Yesterday NVIDIA hosted its keynote as part of its GPU Technology Conference (GTC 2024) providing updates on its accelerated computing products, platforms and solutions across its focus end-markets and the announcement of its next generation Blackwell accelerated GPU computing architecture and next-gen B200 chipset that targets AI/datacenter acceleration. The key takeaways from the event were: 1) The B200 GPU chips show strong performance gains over prior H100 GPU chipset with 2.5x training improvement (FP8) and 5x inference improvement (FP4). 2) An expected move toward a 2-chiplet architecture with 192GB of HBM3e memory. 3) A plethora of Blackwell-based system level solutions (HGX B200, HGX B100, GB200 NVL72) and Grace ARM-based CPU solutions (GB200 - 2 B200 chips and 1 Grace CPU) to enable faster time-to-market for customers and more product segmentation over time. The products will be available to its partners later this year and we believe should drive sustained revenue growth into 2025. 4) The introduction of next gen GPU copper interconnect NVLink 5 across its platforms (similar to H100) and the adoption of 800GB InfiniBand and Ethernet optical networking connectivity (1.6Tbps PAM4 DSP chipset from Marvell) enabling 200Gbps/channel speeds. 5) The team continues to expand its software/ecosystem to accelerate the adoption of AI (especially in enterprise). The team has a multi-pronged strategy in software to accelerate AI adoption through platform partnerships, AI foundry service, and Omniverse Cloud. 6) The team is leveraging its full stack development platform to move into other end markets such as the humanoid robots market. We note that the team has replicated this approach across many end-markets with strong success. Overall, the team is further distancing itself with its aggressive cadence of new product launches and more product segmentation over time. With leading silicon (GPU/DPU/CPU), hardware/software platforms, and a strong ecosystem, NVIDIA is well positioned to continue to benefit from major secular trends in AI, high-performance computing, gaming, and autonomous vehicles, in our view. Bottom line: NVIDIA continues to be 1-2 steps ahead of its competitors. We reiterate our OW rating on NVDA.

- **Remains 1-2 steps ahead of competitors with rollout of Blackwell datacenter GPU architecture (2-chiplet architecture) for next-gen compute acceleration**. During the keynote session, Jen-Hsun Huang, NVIDIA's CEO introduced its leading TSMC 4NP based Blackwell architecture and its B200 datacenter acceleration/GPU silicon product. The new chip has strong performance advantages vs. the prior generation H100 with 2.5x training improvement (FP8) and 5x inference improvement (FP4). Architecturally, B200 is built on TSMC's 4NP process node (208B transistors) with a 2-chiplet architecture. The products will be available to its partners later this year. This will be supported by the 2nd generation transformer engine.

Sources for: Style Exposure – J.P. Morgan Quantitative and Derivatives Strategy; all other tables are company data and J.P. Morgan estimates.

See page 7 for analyst certification and important disclosures.

Harlan Sur AC
(1-415) 315-6700
harlan.sur@jpmorgan.com

Price Performance

	YTD	1m	3m	12m
Abs	78.6%	21.8%	76.6%	243.8%
Rel	70.6%	18.9%	68.0%	213.5%

Company Data

Shares O/S (mn)	2,499
52-week range ($)	974.00-251.30
Market cap ($ mn)	2,210,490.00
Exchange rate	1.00
Free float(%)	96.1%
3M - Avg daily vol (mn)	50.61
3M - Avg daily val ($ mn)	34,631.7
Volatility (90 Day)	44
Index	S&P 500
BBG BUY|HOLD|SELL	60|7|0

Key Metrics (FYE Jan)

$ in millions	FY24A	FY25E
Financial Estimates		
Revenue	60,922	104,142
Adj. EBIT	33,069	63,685
Adj. EBITDA	33,493	64,109
Adj. net income	32,312	56,606
Adj. EPS	12.96	22.70
BBG EPS	24.51	29.17
Cashflow from operations	28,091	48,207
FCFF	27,023	46,907
Margins and Growth		
Revenue Growth Y/Y (%)	125.9%	70.9%
EBIT margin	54.3%	61.2%
EBIT growth	464.5%	92.6%
EBITDA margin	55.0%	61.6%
EBITDA Growth Y/Y (%)	433.2%	91.4%
Net margin	53.0%	54.4%
Adj. EPS growth	288.4%	75.2%
Ratios		
Adj. tax rate	12.1%	17.0%
Interest cover	-	-
Net debt/Equity	NM	NM
Net debt/EBITDA	NM	NM
ROE	99.3%	80.0%
Valuation		
FCFF yield	1.2%	2.1%
Dividend yield	0.1%	0.1%
EV/Revenue	27.4	15.6
EV/EBITDA	49.9	25.3
Adj. P/E	68.3	39.0

Summary Investment Thesis and Valuation

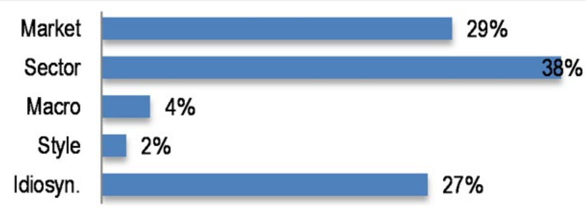
Investment Thesis

We believe NVIDIA continues to execute across all segments. While 1H is typically seasonally weaker than 2H, we expect solid demand in PC gaming to be a strong revenue driver for the company, offsetting PC OEM, which is in secular decline. We expect the data center segment to grow strongly as hyperscale customers continue to embrace GPU-accelerated deep learning for processing large data sets. We are encouraged by strength in the automotive and enterprise segments as well, although strong adoption of autonomous driving in the market remains to be seen. We anticipate significant upside in the shares, driving our Overweight rating.

Valuation

Our Dec-24 PT assumes that NVIDIA trades at 30-35x applied to ~$29.00 of earnings power in CY25, discounted back to CY24 assuming a 12% discount rate. Our outlook of $29.00 of earnings power assumes the team will drive its earnings power by 30%+ per year over the next few years led by the continued strong datacenter growth (30-40% CAGR), monetizing an incremental ~$14B of auto revenue pipeline and an incremental $2-3B of software, licensing, and subscription revenues over the next 3-4 years.

Performance Drivers



Factors	6M Corr	1Y Corr
Market: MSCI US	0.72	0.53
Sect: Technology	0.62	0.76
Ind: Semicond & S Equip	0.78	0.88
Macro:		
Economic Surprise	0.23	0.26
Crude Oil	-0.24	-0.24
US 10yr Breakeven	-0.13	-0.20
Quant Styles:		
DivYld	-0.62	-0.42
Value	-0.39	-0.32
Growth	0.24	0.27

Source: J.P. Morgan Quantitative and Derivatives Strategy for Performance Drivers; company data, Bloomberg Finance L.P. and J.P. Morgan estimates for all other tables. Note: Price history may not be complete or exact.

Harlan Sur AC
(1-415) 315-6700
harlan.sur@jpmorgan.com

- **As expected, NVIDIA also rolled out a plethora of system-level solutions based on the B200 chip architecture alongside CPU and networking solutions… enabling a total system solutions approach and faster time-to-market for customers**. NVIDIA unveiled a plethora of system-level solutions based on the new Blackwell architecture. The team also rolled out is arm-based chip - GB200 superchip which is based on 2 B200 chips and 1 Grace ARM CPU chip. For its system-level solutions, the HGX B200 is based on using eight B200 GPUs. The HGX B100 is the plug and play version for existing HGX H100 infrastructure and allows for rapid deployment. The GB200 NVL72 is the rack scale system (36 GB200 superchips) to support LLMs.
- **Infiniband/Ethernet networking moving to 200Gbps/channel SERDES and ushering in 1.6T optical DSP adoption (Good for Marvell)...continues (similar to H100) to use copper cabling inside of the DGX compute rack with NVLink proprietary connectivity....moving to PCIe Gen6 for within-server connectivity functions...overall powerful connectivity solutions keeping pace with the significant rise of Blackwell's compute power...**Supporting the roll-out of the Blackwell GPU, NVDA has enhanced the chip-to-chip connectivity and overall networking architecture as the data transmission speed has been increased by 2x to 200Gbps per channel (200Gbps SERDES IO speed) and the new NVIDIA Quantum-X800 InfiniBand switch, Spectrum™-X800 Ethernet switch, and its ConnectX®-8 NIC cards support the higher data speeds. Although characterized as 800Gbps Infiniband, the 200Gbps/channel speeds implies that the optical module formfactor will use a 1.6Tbps PAM4 DSP chip (we believe from Marvell). Despite the higher speeds, the team has continued to extend the use of copper connectivity for its proprietary NVLink (GPU-to-GPU) connectivity architecture, which is similar to the H100 within rack connectivity solution. The other big connectivity upgrade is the move to PCIe Gen 6 (CPU to GPU, CPU to storage, CPU to NIC) which doubles the localized data speeds versus the prior generation Gen 5 standard and enables the 200Gbps per channel speeds.
- **Continues to expand its software/ecosystem to accelerate the adoption of AI; also drive high barriers to entry**. NVIDIA's ecosystem and software-driven focus has led and will continue to lead to product leadership across its focus end-markets / domains. The team announced strong partnerships with the EDA and systems analysis vendors (e.g. SNPS, CDNS, and ANSS). In software, the team has a multi-prong strategy to drive strong AI adoption. The team introduced NVIDIA Inference Microservice (NIM), pre-trained models that are packaged and optimized to run across NVIDIA's installed base. The team is accelerating adoption of its AI foundry service to enable AI usage within enterprise software and data platforms such as SAP, ServiceNow, Cohesity, and Snowflake. The team showed customer traction with Omniverse Cloud (e.g. Siemens and Nissan) and announced that Omniverse Cloud streams to Apple's Vision Pro headset. Software revenues are at a $1B revenue run-rate and we believe its a multi-billion dollar revenue opportunity and is annuity-like and a very high margin revenue stream.
- **Leveraging full stack development platform to move into the humanoid robots market.** A significant part of NVIDIA's overall success is the company's realization that leadership is driven by a combination of having the proper ecosystem (silicon, hardware, systems and software) and fostering strong industry relationships/ partnership. The team has replicated this success across many end markets and we believe will be key drivers in their ability to address the humanoid robots market. The team is leveraging its Project GR00T foundation model (software to train/teach robots) and Jetson Thor robotics computer (based on Blackwell). In automotive, the

Harlan Sur [AC]
(1-415) 315-6700
harlan.sur@jpmorgan.com

team continues to expand it automotive pipeline and announced BYD will be adopting DRIVE Thor.

Harlan Sur [AC]
(1-415) 315-6700
harlan.sur@jpmorgan.com

Investment Thesis, Valuation and Risks

NVIDIA Corporation *(Overweight; Price Target: $850.00)*

Investment Thesis

We believe NVIDIA continues to execute across all segments. While 1H is typically seasonally weaker than 2H, we expect solid demand in PC gaming to be a strong revenue driver for the company, offsetting PC OEM, which is in secular decline. We expect the data center segment to grow strongly as hyperscale customers continue to embrace GPU-accelerated deep learning for processing large data sets. We are encouraged by strength in the automotive and enterprise segments as well, although strong adoption of autonomous driving in the market remains to be seen. We anticipate significant upside in shares, driving our Overweight rating.

Valuation

Our Dec-24 PT assumes that NVIDIA trades at 30-35x applied to ~$29.00 of earnings power in CY25E, discounted back to CY24 assuming a 12% discount rate. Our outlook of $29.00 of earnings power assumes the team will drive its earnings power by 30%+ per year over the next few years led by the continued strong datacenter growth (30-40% CAGR), monetizing an incremental ~$14B of auto revenue pipeline and an incremental $1-2B of software, licensing, and subscription revenues over the next 3-4 years.

Risks to Rating and Price Target

- Although PC gaming demand seems resilient to macroeconomic weakness, any macro uncertainty could weigh on PC gaming demand trends. Given NVDA's ~53% exposure to the PC gaming segment, any consumer PC gaming weakness poses downside risk to our estimates.
- NVIDIA's GPUs could gain lower than expected deployment into data center applications as hyperscale customers further adopt deep learning as a new and effective way of processing large unstructured data sets. Any significant decrease in the adoption of deep learning by hyperscale customers or increase in competition could result in downside risk to our revenue and earnings estimates.

Harlan Sur [AC]
(1-415) 315-6700
harlan.sur@jpmorgan.com

NVIDIA Corporation: Summary of Financials

Income Statement - Annual	FY23A	FY24A	FY25E	FY26E	FY27E
Revenue	**26,974**	**60,922**	**104,142**	**-**	**-**
COGS	(11,618)	(16,621)	(26,093)	-	-
Gross profit	**15,356**	**44,301**	**78,050**	**-**	**-**
SG&A	(2,160)	(2,558)	(3,036)	-	-
Adj. EBITDA	**6,282**	**33,493**	**64,109**	**-**	**-**
D&A	(424)	(424)	(424)	-	-
Adj. EBIT	**5,858**	**33,069**	**63,685**	**-**	**-**
Net Interest	-	-	-	-	-
Adj. PBT	**5,864**	**33,680**	**64,685**	**-**	**-**
Tax	186	(4,059)	(10,997)	-	-
Minority Interest	-	-	-	-	-
Adj. Net Income	**8,365**	**32,312**	**56,606**	**-**	**-**
Reported EPS	1.74	11.94	21.34	-	-
Adj. EPS	**3.34**	**12.96**	**22.70**	**-**	**-**
DPS	**0.68**	**0.68**	**0.68**	**-**	**-**
Payout ratio	39.0%	5.7%	3.2%	-	-
Shares outstanding	2,507	2,493	2,493	-	-

Balance Sheet & Cash Flow Statement	FY23A	FY24A	FY25E	FY26E	FY27E
Cash and cash equivalents	13,296	25,984	72,495	-	-
Accounts receivable	3,827	9,999	14,598	-	-
Inventories	5,159	5,282	9,945	-	-
Other current assets	791	3,080	3,080	-	-
Current assets	**23,073**	**44,345**	**100,118**	**-**	**-**
PP&E	3,807	3,914	4,790	-	-
LT investments	-	-	-	-	-
Other non current assets	14,302	17,469	17,469	-	-
Total assets	**41,182**	**65,728**	**122,377**	**-**	**-**
Short term borrowings	1,250	1,250	1,250	-	-
Payables	1,193	2,699	3,807	-	-
Other short term liabilities	4,120	6,682	6,682	-	-
Current liabilities	**6,563**	**10,631**	**11,739**	**-**	**-**
Long-term debt	9,703	8,459	8,459	-	-
Other long term liabilities	2,815	3,660	3,660	-	-
Total liabilities	**19,081**	**22,750**	**23,858**	**-**	**-**
Shareholders' equity	22,101	42,978	98,520	-	-
Minority interests	-	-	-	-	-
Total liabilities & equity	**41,182**	**65,728**	**122,377**	**-**	**-**
BVPS	**8.81**	**17.24**	**39.51**	**-**	**-**
y/y Growth	*(16.0%)*	*95.6%*	*129.2%*	-	-
Net debt/(cash)	(2,343)	(16,275)	(62,786)	-	-
Cash flow from operating activities	**5,642**	**28,091**	**48,207**	**-**	**-**
o/w Depreciation & amortization	424	424	424	-	-
o/w Changes in working capital	(3,174)	(5,057)	(10,370)	-	-
Cash flow from investing activities	**7,374**	**(10,566)**	**(1,300)**	**-**	**-**
o/w Capital expenditure	(1,833)	(1,068)	(1,300)	-	-
as % of sales	*6.8%*	*1.8%*	*1.2%*	-	-
Cash flow from financing activities	**(11,617)**	**(13,634)**	**(396)**	**-**	**-**
o/w Dividends paid	(398)	(396)	(396)	-	-
o/w Net debt issued/(repaid)	(7)	(1,279)	0	-	-
Net change in cash	**1,399**	**3,891**	**46,511**	**-**	**-**
Adj. Free cash flow to firm	**3,809**	**27,023**	**46,907**	**-**	**-**
y/y Growth	*(53.2%)*	*609.5%*	*73.6%*	-	-

Income Statement - Quarterly	1Q25E	2Q25E	3Q25E	4Q25E
Revenue	**24,000**	**25,461**	**26,693**	**27,988**
COGS	(5,688)	(6,543)	(6,780)	(7,081)
Gross profit	**18,312**	**18,917**	**19,913**	**20,907**
SG&A	(731)	(746)	(761)	(799)
Adj. EBITDA	**14,939**	**15,474**	**16,399**	**17,297**
D&A	(106)	(106)	(106)	(106)
Adj. EBIT	**14,833**	**15,368**	**16,293**	**17,191**
Net Interest	-	-	-	-
Adj. PBT	**15,083**	**15,618**	**16,543**	**17,441**
Tax	(2,564)	(2,655)	(2,812)	(2,965)
Minority Interest	-	-	-	-
Adj. Net Income	**13,248**	**13,692**	**14,460**	**15,206**
Reported EPS	4.98	5.15	5.46	5.75
Adj. EPS	**5.32**	**5.49**	**5.80**	**6.09**
DPS	**0.17**	**0.17**	**0.17**	**0.17**
Payout ratio	3.4%	3.3%	3.1%	3.0%
Shares outstanding	2,491	2,492	2,494	2,496

Ratio Analysis	FY23A	FY24A	FY25E	FY26E	FY27E
Gross margin	56.9%	72.7%	74.9%	-	-
EBITDA margin	23.3%	55.0%	61.6%	-	-
EBIT margin	21.7%	54.3%	61.2%	-	-
Net profit margin	31.0%	53.0%	54.4%	-	-
ROE	34.3%	99.3%	80.0%	-	-
ROA	19.6%	60.4%	60.2%	-	-
ROCE	16.1%	67.8%	65.7%	-	-
SG&A/Sales	8.0%	4.2%	2.9%	-	-
Net debt/equity	NM	NM	NM	-	-
P/E (x)	265.1	68.3	39.0	-	-
P/BV (x)	100.3	51.3	22.4	-	-
EV/EBITDA (x)	268.0	49.9	25.3	-	-
Dividend Yield	0.1%	0.1%	0.1%	-	-
Sales/Assets (x)	0.6	1.1	1.1	-	-
Interest cover (x)	-	-	-	-	-
Operating leverage	(19426.5%)	369.1%	130.5%	-	-
Revenue y/y Growth	0.2%	125.9%	70.9%	-	-
EBITDA y/y Growth	(41.6%)	433.2%	91.4%	-	-
Tax rate	3.2%	12.1%	17.0%	-	-
Adj. Net Income y/y Growth	(25.7%)	286.3%	75.2%	-	-
EPS y/y Growth	(24.9%)	288.4%	75.2%	-	-
DPS y/y Growth	0.0%	0.0%	0.0%	-	-

Source: Company reports and J.P. Morgan estimates.
Note: $ in millions (except per-share data).Fiscal year ends Jan. o/w - out of which

Harlan Sur [AC]
(1-415) 315-6700
harlan.sur@jpmorgan.com

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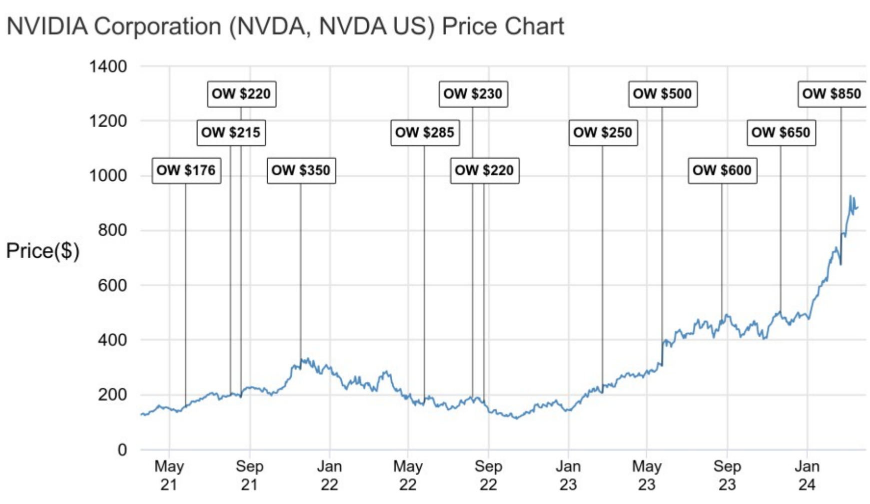

Date	Rating	Price ($)	Price Target ($)
27-May-21	OW	157.00	176
03-Aug-21	OW	197.50	215
19-Aug-21	OW	190.40	220
18-Nov-21	OW	292.61	350
26-May-22	OW	169.75	285
08-Aug-22	OW	189.89	230
25-Aug-22	OW	172.22	220
23-Feb-23	OW	207.54	250
25-May-23	OW	305.38	500
24-Aug-23	OW	471.16	600
22-Nov-23	OW	499.44	650
22-Feb-24	OW	674.72	850

Source: Bloomberg Finance L.P. and J.P. Morgan; price data adjusted for stock splits and dividends.
Initiated coverage Apr 04, 2002. All share prices are as of market close on the previous business day.

The chart(s) show J.P. Morgan's continuing coverage of the stocks; the current analysts may or may not have covered it over the entire period.
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	Overweight (buy)	Neutral (hold)	Underweight (sell)
J.P. Morgan Global Equity Research Coverage*	47%	39%	13%
IB clients**	48%	43%	32%
JPMS Equity Research Coverage*	46%	42%	12%
IB clients**	68%	63%	46%

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Harlan Sur [AC]
(1-415) 315-6700
harlan.sur@jpmorgan.com

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Harlan Sur [AC]
(1-415) 315-6700
harlan.sur@jpmorgan.com

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